

EDRM 718 Adam Brown Final Project

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Description

This research report explores the relationships between optimism scores (from the survey data and measures their outlook on employment and career readiness), reported GPAs, and the college students academic disciplines or subjects. More specifically, this report seeks to answer these three questions, Is senior-year grade point average (GPA) related to the degree of optimism about future employment? Is the relationship of GPA and optimism consistent or different for different categories of disciplines? Is the level of optimism different for different categories of disciplines? The three datasets include survey results from over 700 college seniors. graduating this spring, representing 23 major universities in the United States. These datasets include demographic data, academic performance data, and data from a short (14-item) Likert-type survey which sought to determine their optimism regarding their readiness and likelihood of obtaining a job after graduation

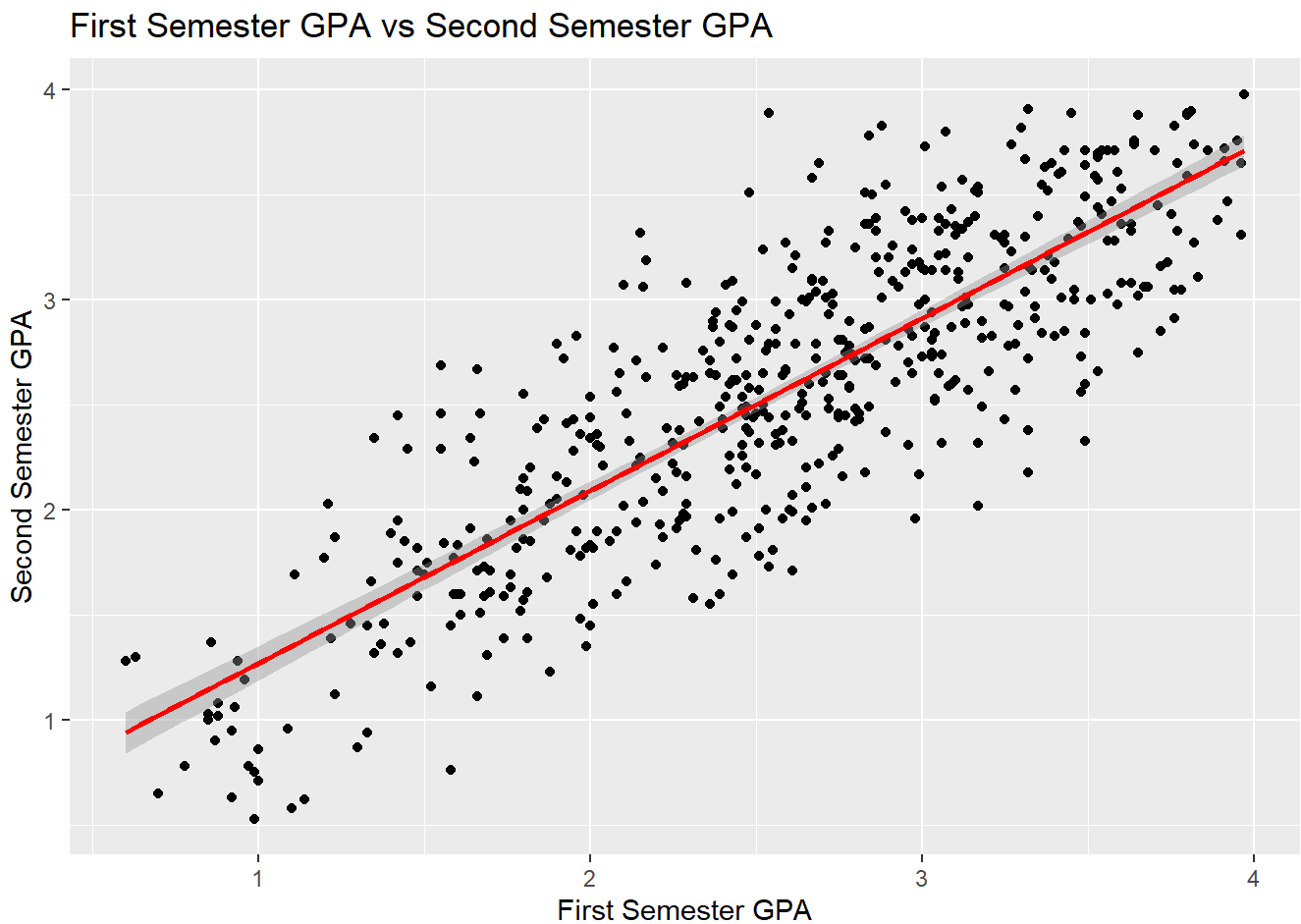
Source of Data: Real World Solutions, Job Prep Demographics, Job Prep GPA, and Job Prep Survey Results. Sourced 24 April 2025.

Analysis

I begin my analysis reading in the three separate data files, tidying the data that is missing or has problems, and then joining the files together using my unique key, which in this case is Student ID number. Now that I have completed the data processing and cleaning, I start by examining the reported GPAs for the students from the first and second semester. We need to verify consistency of GPAs before we can create a weighted GPA average.

In Figure 1, I created a scatterplot to verify consistency by observing if there a strong linear relationship exists between first and second semester GPAs. In Figure 1, we observe that as the GPA for the first semester increases, so does the GPA for the second semester. This tell us that a strong linear relationship does exist. Since a strong linear relationship exists this GPAs are comparable which means we can create our next scatterplot which will helps us to know if we can predict second semester GPA from first semester GPA.

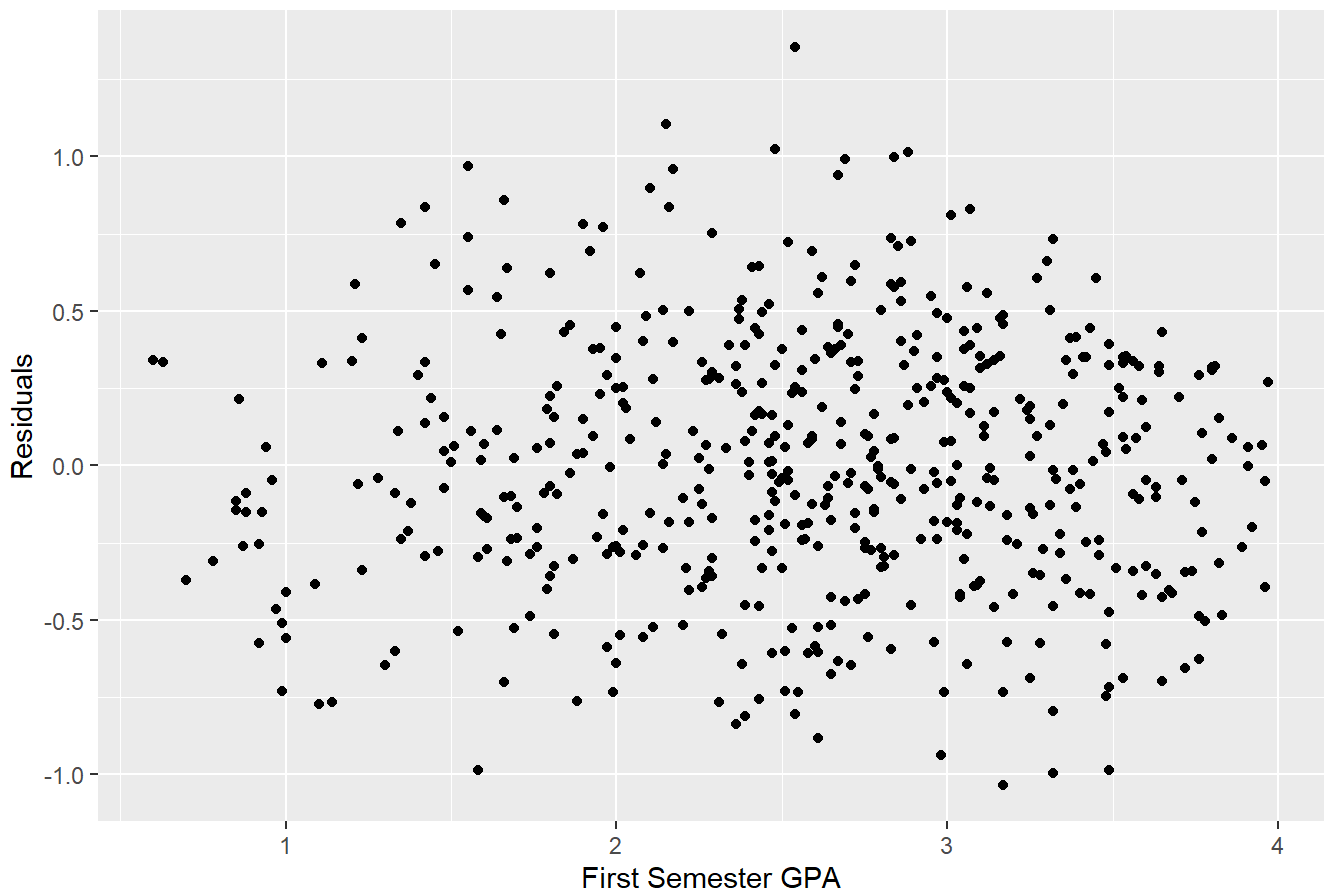
Figure 1. Scatterplot of First Semester GPA Vs Second Semester GPA.



In Figure 2, I created a scatterplot to identify if patterns exist or not. We do not observe any shapes or patterns in the data, which is good because this means the residuals are randomly distributed. This scatterplot helps me to know that my linear model is a good fit and there shouldn't be significant bias in predicting second semester GPA from first semester. This gives me confidence to create a weighted GPA but first I need to identify and exclude those students who are among those with the ten percent largest residuals. I run this analysis next in Figure 3.

Figure 2. Scatterplot of Residuals from Predicted Second Semester GPA.

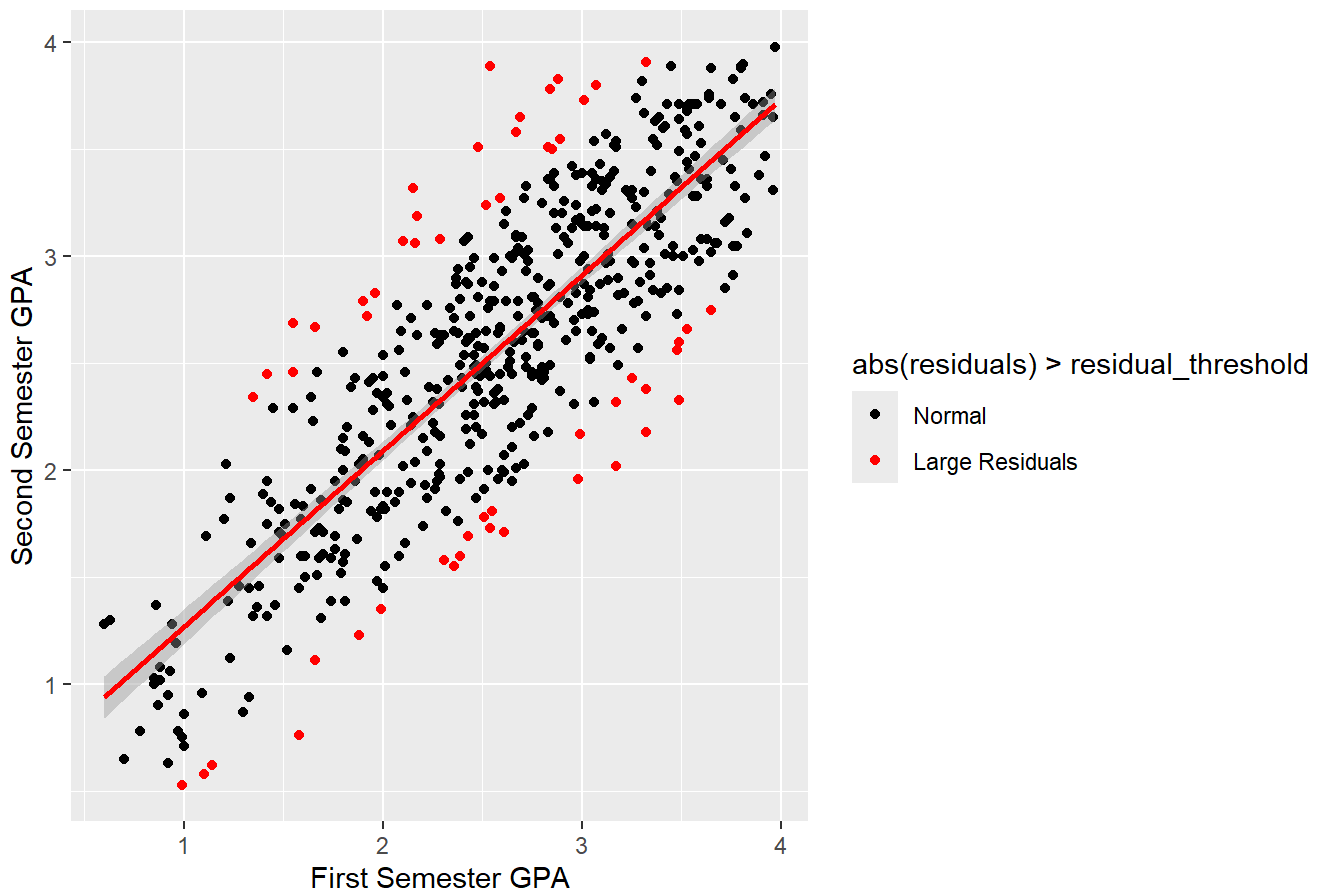
Residuals from Predicted Second Semester GPA



In Figure 3, we can observe the students who are among those with the ten percent largest residuals as they are noted in red. We note from our data that there are 54 students who fall into this category. I will remove those students from the data prior to calculating weighted GPA, which we will call "Senior Year GPA" and we will continue to move through the analysis.

Figure 3. Scatterplot of First Semester GPA Vs Second Semester GPA with Large Residuals.

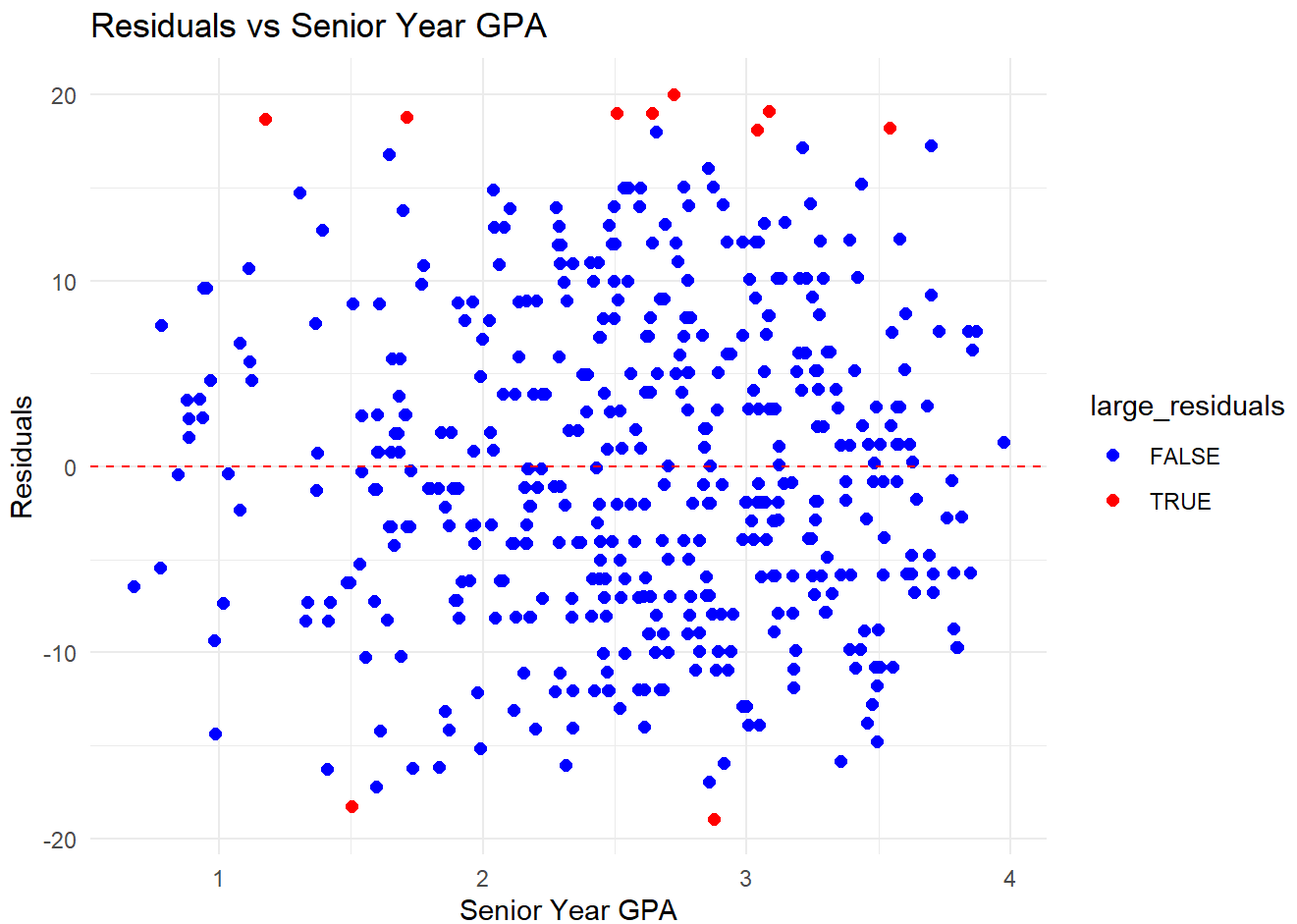
First Semester GPA vs Second Semester GPA with Large Residuals Highlighted



Analysis for Question #1: Is Senior-year Grade Point Average (GPA) Related to the Degree of Optimism about Future Employment?

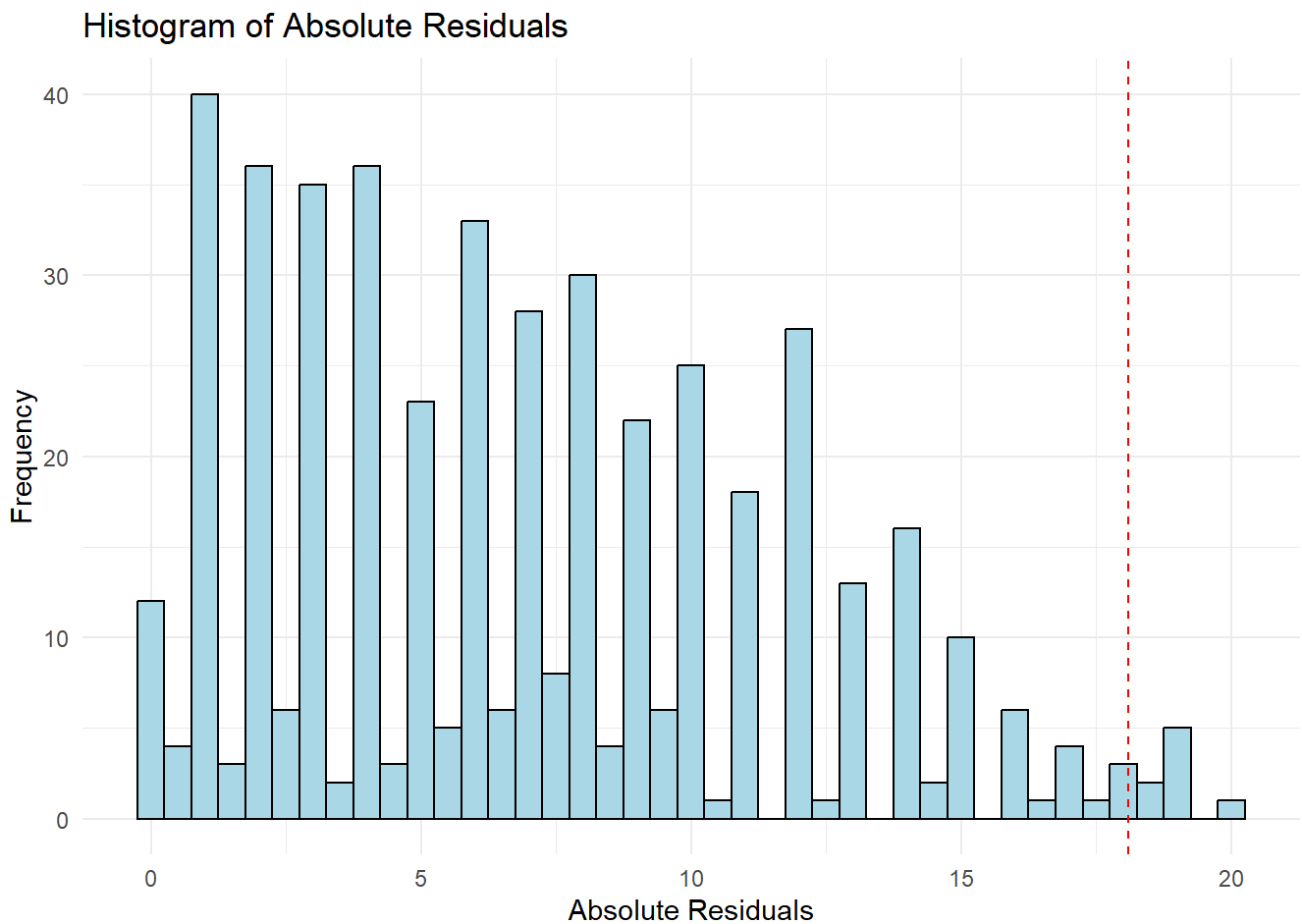
In figure 4, we start to answer our first research question, Is senior-year grade point average (GPA) related to the degree of optimism about future employment? We start by verifying consistency by reviewing the ten students with the largest residuals in the dataset. We can observe these students by the red dots in Figure 4. We analyze these residuals further in Figure 5.

Figure 4. Scatterplot of Residuals of Senior Year GPA Vs Optimism Score.



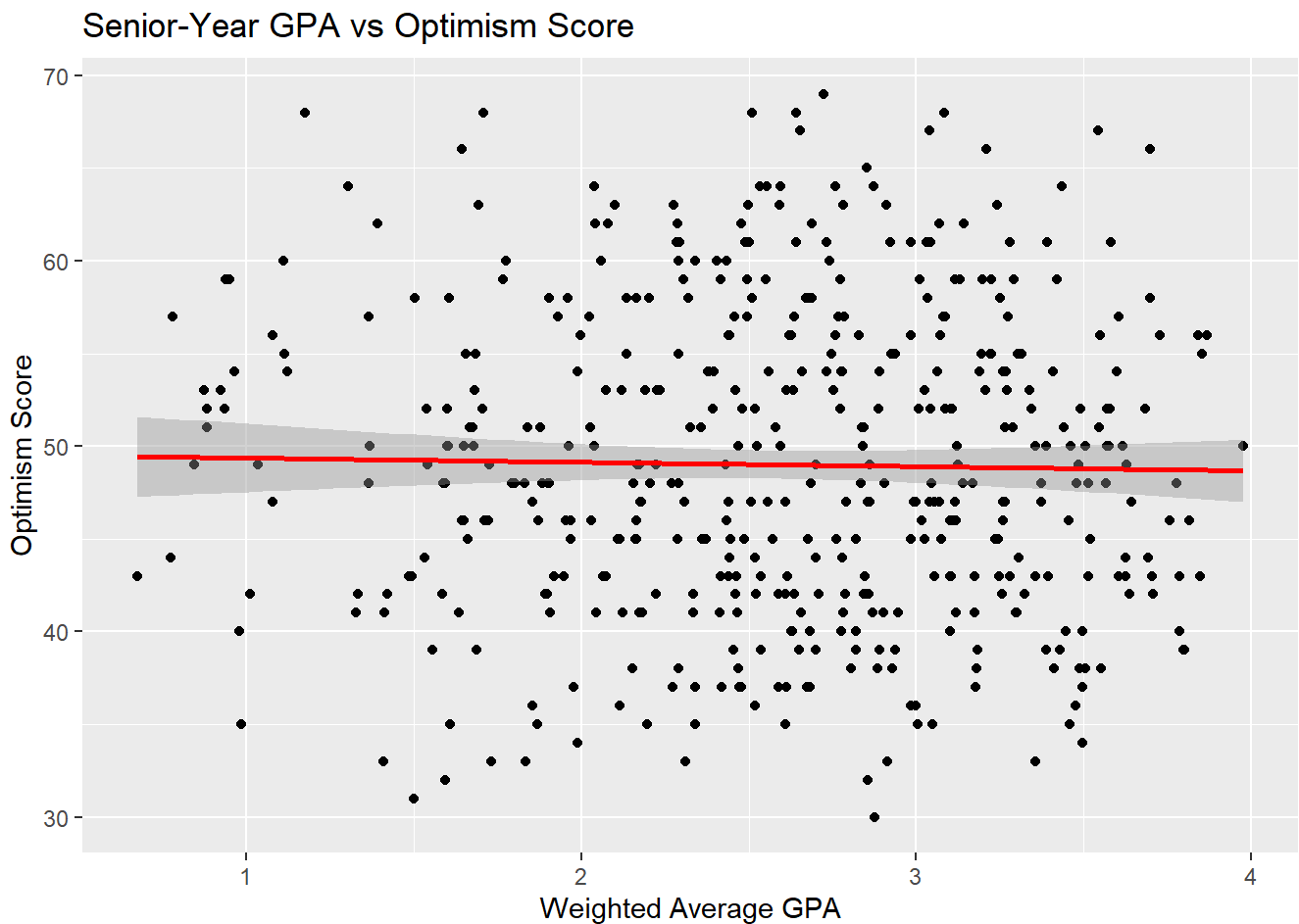
In Figure 5, we further explore the residuals to ensure consistency prior to running more analysis. Again, we observe a similar trend in the data in the Histogram that we did in Figure 4. There are a few students who have higher residuals, however, after reviewing these students in the dataset, they show no discernible pattern between them on Subjects, GPA, Optimism Score, etc. So, it will not strengthen our model to remove these students, thus we retain them in the dataset.

Figure 5. Histogram of the Frequency of Absolute Residuals.



In Figure 6, we observe the scatterplot of Senior Year GPA vs Optimism Score. From our observations, it appears there is not a strong relationship between the two variables. Meaning that as GPA increases, optimism scores does not necessarily increase or decrease. We examine these variables further in Table 1.

Figure 6. Scatterplot of Senior Year GPA (Weighted GPA) Vs Optimism Score.



In Table 1, we observe the correlation coefficient between Senior Year GPA and Optimism score. , Based on the coefficient, there is a very small and negative relationship between these two variables. As we predicted from Figure 6, no strong relationship exists. To be sure, we run one more analysis in Table 2.

Table 1. Correlation Between Senior Year GPA and Optimism Score.

Correlation Between Weighted GPA and Optimism Score		
Variable 1	Variable 2	Correlation_Coefficient
weighted_avg_gpa	total_score	-0.02

In Table 2, we again learn that there is a small negative relationship between the two variables and this helps us answer no to the question: Is senior-year grade point average (GPA) related to the degree of optimism about future employment?

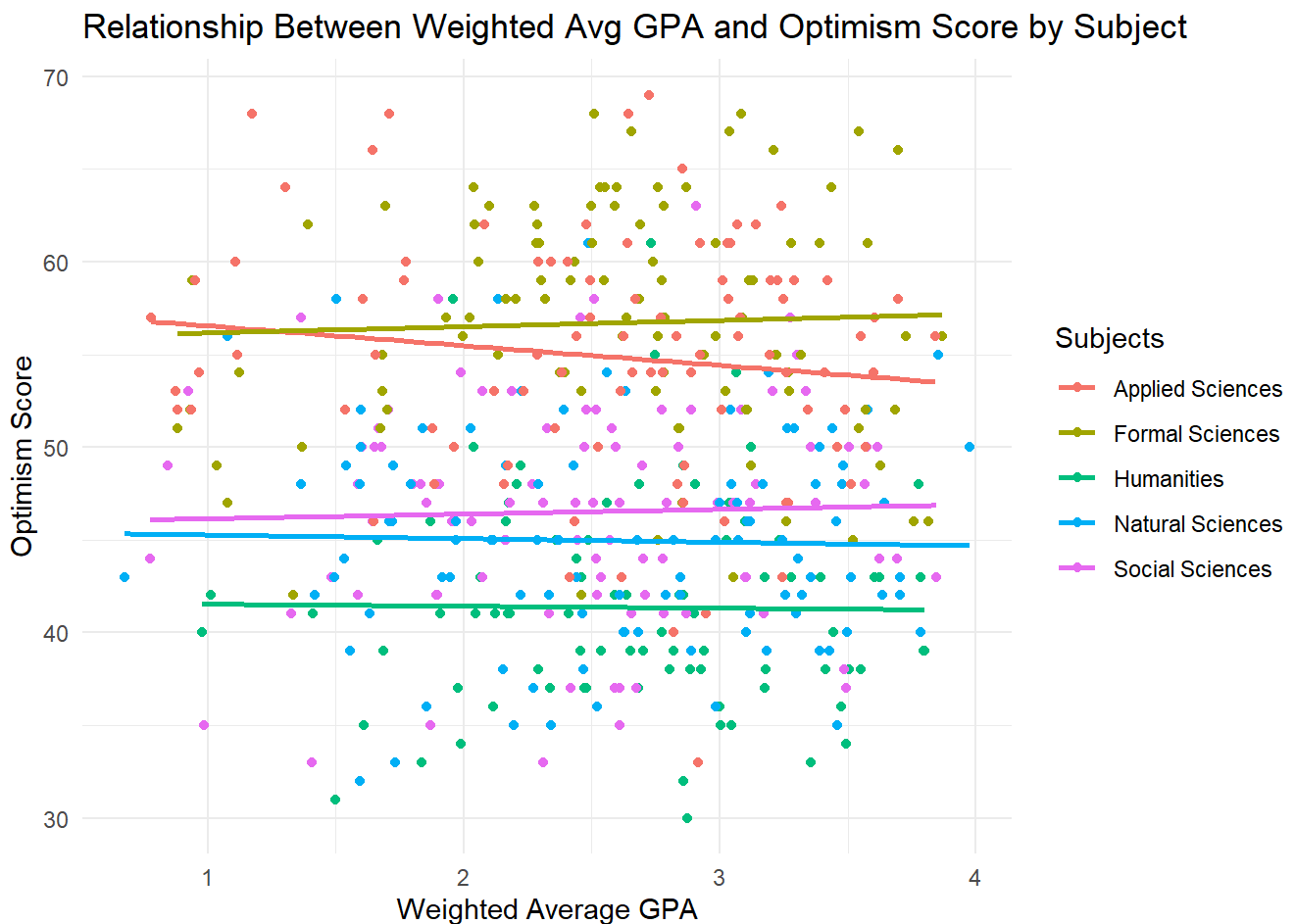
Table 2. Linear Regression of Optimism Score and Senior Year GPA.

Linear Regression Results: Optimism Score vs Weighted GPA				
Term	Estimate	Standard Error	t-Statistic	p-Value
(Intercept)	49.60	1.45	34.12	0.00
weighted_avg_gpa	-0.23	0.54	-0.42	0.67

Analysis for Question #2: Is the Relationship of GPA and Optimism Consistent or Different for Different Categories of Disciplines?

In figure 6, we observe the relationship between Senior Year GPA and Optimism Score is not consistent across different subjects or disciplines. For example, we note that they relationship between Senior Year GPA and Optimism Score in some subjects, like Humanities and Natural Sciences, appears to be negative. However, subjects like Applied Sciences and Social Sciences appear to have a weak or negative relationship with Optimism Score. To be more conclusive, we examine this question further in Table 3 with Linear Regression of Subjects.

Figure 6. Scatterplot of Senior Year GPA Vs Optimism Score by Subject.



In Table 3, we further seek to answer the question, Is the relationship of GPA and optimism consistent or different for different categories of disciplines? We learn from the linear regression in Table 3, with either negative or small coefficients for each subject, that the relationship must in fact be different for different categories of disciplines. However, we run one more test to be sure in Table 4, an interaction model between Optimism Score and Senior Year GPA by Subject.

Table 3. Linear Regression of Subjects.

Linear Regression Model Results by Subject			
Subject	Coefficient of Senior Year GPA		p-value
Applied Sciences	-1.06		0.24
Formal Sciences	0.34		0.70
Humanities	-0.11		0.91
Natural Sciences	-0.19		0.80
Social Sciences	0.25		0.77

In table 4, we observe the interaction model (Relationship) of Optimism Score and Senior Year GPA across different subjects. We observe, like in Figure 6, the relationship is indeed different across different subjects and mostly negative when comparing each subject to Applied Sciences for Optimism Score. Meaning that in general as GPA increases and is held constant, Optimism Scores decreases. However, based on the interaction data with the subjects in Table 4, we learn that there is a positive relationship for all subjects when compared to Applied Sciences. This tells us that the interaction between Senior Year GPA and Optimism Score is stronger in all subjects compared to Applied Sciences. The good news that from Table 4 we can interpret the confidence intervals and determine with 95% confidence where we would expect a score to be. For example, we can predict with 95% confidence that when GPA is held constant, an optimism score for Social Sciences will fall between -17.95 and -5.48. This confidence interval does not include zero, meaning it is statistically significant. However, for the confidence intervals that do include zero, these would not be considered statically significant. In summary, Is the relationship of GPA and optimism consistent or different for different categories of disciplines? Yes, it is different for different categories of disciplines.

Table 4. Interaction Model of Optimism Score Vs Senior Year GPA, and Subjects.

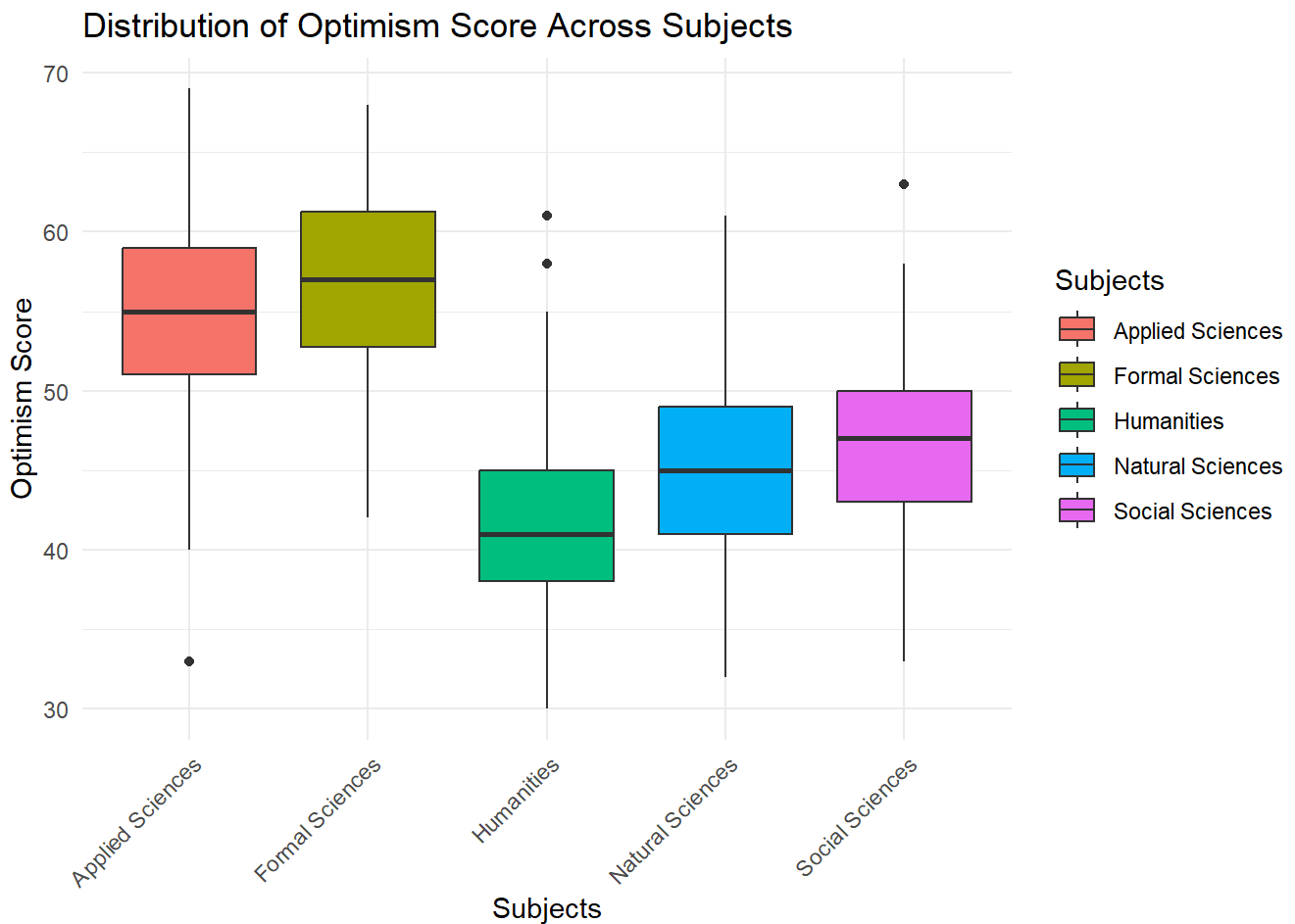
Interaction Model Results: Optimism Score vs Senior Year GPA and Subjects						
Term	Estimate	Standard Error	t-Statistic	p-Value	95% CI Lower	95% CI Upper
(Intercept)	57.61	2.21	26.11	0.00	53.27	61.95

Interaction Model Results: Optimism Score vs Senior Year GPA and Subjects						
Term	Estimate	Standard Error	t-Statistic	p-Value	95% CI Lower	95% CI Upper
weighted_avg_gpa	-1.06	0.83	-1.28	0.20	-2.69	0.56
SubjectsFormal Sciences	-1.78	3.21	-0.55	0.58	-8.08	4.52
SubjectsHumanities	-15.97	3.58	-4.46	0.00	-23.00	-8.94
SubjectsNatural Sciences	-12.15	3.13	-3.89	0.00	-18.29	-6.00
SubjectsSocial Sciences	-11.72	3.17	-3.69	0.00	-17.95	-5.48
weighted_avg_gpa:SubjectsFormal Sciences	1.40	1.19	1.18	0.24	-0.94	3.74
weighted_avg_gpa:SubjectsHumanities	0.96	1.32	0.72	0.47	-1.65	3.56
weighted_avg_gpa:SubjectsNatural Sciences	0.87	1.16	0.75	0.46	-1.42	3.16
weighted_avg_gpa:SubjectsSocial Sciences	1.32	1.21	1.08	0.28	-1.07	3.70

Analysis for Question #3: Is the Level of Optimism Different for Different Categories of Disciplines?

In Figure 8 we can observe the following in relationship to our question, Is the level of optimism different for different categories of disciplines?, students in applied and formal sciences appear to have higher and more consistent levels of optimism when compared with students in humanities or natural sciences. In addition, students in the social sciences discipline have higher variability than other disciplines, meaning some student are very optimistic while others are very pessimistic. I will run more analysis on this data in Table 5 by running an ANOVA test.

Figure 8. Distribution of Optimism Scores Across Subjects.



In table 5, we observe a one-way ANOVA test of Optimism Scores across different subjects to help us continue to answer our research question. In this table we observe an F-value of 114.10 which is quite large and tells us there is a strong relationship between Subjects and Optimism Scores and that the level of optimism about future employment differs significantly based on the subject. The observed P value also tell us that this could be statistically significant, and we would normally reject the null hypothesis of: there is no observed difference in optimism scores across subjects. However, in the next table we run 95% confidence intervals which will tell us with greater confidence.

Table 5. One-way ANOVA of Optimism Score Across Subjects.

ANOVA Results for Optimism Score by Subject					
Degrees of Freedom	Sum of Squares	Mean Squares	F Value	p-value	
4	16,558.74	4,139.69	114.10	0.00	
473	17,161.25	36.28	NA	NA	

In Table 6, we observe in general by reviewing the results of the table that the level of optimism is significantly different across various subjects. More specifically, Humanities has a much lower level of optimism than that of all of other sciences. In addition, the differences between Applied, Formal, and Natural Sciences are significant in most cases, while the comparison between Social and Natural Sciences is

not significant. For example, we can predict with 95% confidence that the optimism score for Humanities will fall between -17.77 and -12.95 lower than Formal Sciences, with GPA held constant. Finally, we can answer our research question, Is the level of optimism different for different categories of disciplines? And the answer is Yes.

Table 6. Tukey Post-hoc Test of ANOVA Results in Table 5.

Tukey's HSD Post-Hoc Test Results				
Comparison of Subjects	Difference in Means	Lower Bound of CI	Upper Bound of CI	p-value
Formal Sciences-Applied Sciences	1.83	-0.55	4.20	0.22
Humanities-Applied Sciences	-13.53	-15.98	-11.08	0.00
Natural Sciences-Applied Sciences	-9.93	-12.30	-7.56	0.00
Social Sciences-Applied Sciences	-8.37	-10.77	-5.97	0.00
Humanities-Formal Sciences	-15.36	-17.77	-12.95	0.00
Natural Sciences-Formal Sciences	-11.76	-14.09	-9.43	0.00
Social Sciences-Formal Sciences	-10.20	-12.56	-7.84	0.00
Natural Sciences-Humanities	3.60	1.19	6.00	0.00
Social Sciences-Humanities	5.16	2.72	7.59	0.00
Social Sciences-Natural Sciences	1.56	-0.79	3.91	0.36

Conclusion

This report and analysis aimed to reveal the answers to three research questions, Is senior-year grade point average (GPA) related to the degree of optimism about future employment? Is the relationship of GPA and optimism consistent or different for different categories of disciplines? Is the level of optimism different for different categories of disciplines?, which have been answered through the analysis. First, the analysis revealed that there is not a statistically significant relationship between Senior Year GPA and the degree of

optimism about future employment. Meaning that Senior Year GPA does not have a meaningful influence on optimism about future employment. Next, the relationship between GPA and optimism does not appear to be consistent across different academic disciplines. The research also suggested that the impact of Senior Year GPA on Optimism Score depends on the Subject. Lastly, based on the data presented, the level of optimism about future employment does differ significantly across Subjects.